1. Indicator 2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

(Judge the reasonableness of a conclusion in context)

- Highly relevant (⭐⭐⭐⭐⭐)
  This is the central focus of the task. Kevin's reasoning appears plausible at first glance—both countries have similar TV ownership rates (~97%). However, he overlooks a crucial fact: France has far more TV-owning households in absolute terms (24.5 million) than Norway (2.0 million).
  Although Norway has a higher cable subscription rate among TV-owning households (42.7% vs. 15.4%), the actual number of subscribers is:

  - Norway: $2.0 \times 0.427 \approx 0.854$ million
  - France: $24.5 \times 0.154 \approx 3.773$ million

  Clearly, France has significantly more cable subscribers. Kevin's error stems from confusing proportion with absolute quantity, and students must recognize that his conclusion is unreasonable in the given context.

---

2. Indicator 7: Construct and communicate explanations and arguments in the context of the problem

(Build and articulate reasoned justifications)

- Highly relevant (⭐⭐⭐⭐☆)
  The prompt explicitly asks: *"Explain why this statement is incorrect. Give a reason."* Students cannot merely spot the flaw—they must formulate a clear, logical explanation, such as:

  > "Although TV ownership rates are similar, France has many more TV-owning households overall. Even with a lower cable subscription rate, the total number of cable subscribers in France is much larger than in Norway."
  > This reflects higher-order thinking: moving from error detection to constructing a valid counterargument.

3. Indicator 1: Interpret a mathematical result back into the real-world context

(Reconnect numerical results to real-life meaning)

- Highly relevant (⭐⭐⭐⭐☆)
  Students must translate the tabular figures—24.5M vs. 2.0M households, 15.4% vs. 42.7% subscription rates—into an understanding of real-world scale: France is a much larger country in population and household count. Even a smaller percentage of a large base yields a greater absolute number. This contextual reinterpretation is essential for a correct response.

---

4. Indicator 9: Understand the relationship between the problem context and the mathematical representation... and gauge the feasibility of the solution

(Link context to data representation and assess plausibility)

- Moderately high relevance (⭐⭐⭐☆☆)
  The data are presented in a hierarchical percentage structure: cable subscription rates are given as percentages *of TV-owning households*, not of all households. Kevin likely misinterpreted this layered representation. Correctly understanding how the data are structured is key to evaluating the feasibility of his claim.

---

5. Indicator 6: Use knowledge of how the real world impacts the outcomes... to make contextual judgments

(Apply real-world knowledge to interpret results)

- Moderate relevance (⭐⭐⭐☆☆)
  Students can draw on general knowledge: France (~60 million people) is vastly more populous than Norway (~5 million), so it naturally has far more households. This background awareness helps them question Kevin's conclusion even without precise computation.

6. Indicator 5: Interpret mathematical outcomes in various formats… compare or evaluate multiple representations

(Integrate and compare different types of quantitative information)

- Moderate relevance (⭐⭐⭐☆☆)
  The task requires students to simultaneously compare three types of data across two countries:
  - Absolute number of TV-owning households
  - TV ownership rate (% of all households)
  - Cable subscription rate (% of TV-owning households)
    They must recognize that total cable subscribers = (TV households) × (subscription rate)—a synthesis of absolute and relative data.

1. 第2条：Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

（判断某个数学结论在问题情境下是否合理）

- 高度相关（⭐⭐⭐⭐⭐）
  这是本题的核心考查点。Kevin 的结论看似合理（两国电视普及率相近 ≈97%），但他忽略了关键事实：法国拥有电视的家庭总数（24.5 million）远高于挪威（2.0 million）。
  尽管挪威有更高比例的电视家庭订阅有线电视（42.7% vs. 15.4%），但其绝对订阅户数为：
  - 挪威：2.0 × 42.7% ≈ 0.854 million
  - 法国：24.5 × 15.4% ≈ 3.773 million
    显然，法国的有线电视订阅户数远多于挪威。Kevin 的错误在于混淆了“比例”与“绝对数量”，而学生必须识别这一推理在现实情境中的不合理性。

---

2. 第7条：Construct and communicate explanations and arguments in the context of the problem

（在问题情境中构建并交流解释与论证）

- 高度相关（⭐⭐⭐⭐☆）
  题目明确要求“Explain why this statement is incorrect. Give a reason.” 学生不能仅指出错误，还需用清晰、合乎逻辑的语言说明原因，例如：

  > “Although the TV ownership rates are similar, France has far more TV-owning households in total. Even with a lower cable subscription rate, the absolute number of cable subscribers in France is much larger than in Norway.”
  > 这体现了从“识别错误”到“构建有效反驳论证”的高阶思维。

---

3. 第1条：Interpret a mathematical result back into the real world context

（将数学结果重新解释回现实世界情境）

- 高度相关（⭐⭐⭐⭐☆）
  学生需将表格中的数字（24.5M vs. 2.0M；15.4% vs. 42.7%）转化为对现实规模的理解：法国是一个人口大国，其家庭基数远超挪威。即使订阅比例较低，其总量仍占优。这种“从数据到现实意义”的转换是正确解释的基础。

4. 第9条：Understand the relationship between the context of the problem and representation of the mathematical solution… gauge the feasibility… of the solution

（理解问题情境与数学表示之间的关系，评估解的可行性）

- 中高相关（⭐⭐⭐☆☆）
  表格以复合百分比结构呈现数据（% of all households → % of TV-owning households）。学生必须理解这种层级关系：有线电视订阅比例是以“拥有电视的家庭”为分母，而非全国总户数。Kevin 的错误部分源于对数据表示结构的误解。正确理解这一表示方式是判断其结论可行性的前提。

---

5. 第6条：Use knowledge of how the real world impacts the outcomes… to make contextual judgments

（利用现实世界知识对结果做出情境化判断）

- 中等相关（⭐⭐⭐☆☆）
  学生可借助常识：法国人口（约6000万）远大于挪威（约500万），因此家庭总数也应更大。这种背景知识支持他们质疑 Kevin 的结论，即使不进行精确计算也能推断法国的绝对订阅数可能更高。

---

6. 第5条：Interpret mathematical outcomes in a variety of formats… compare or evaluate two or more representations

（解读不同格式的数学结果，比较多个表示）

- 中等相关（⭐⭐⭐☆☆）
  题目要求学生同时比较两个国家的三类数据：
  - 电视家庭总数（绝对数）
  - 电视普及率（相对比例）
  - 有线电视订阅率（条件比例）
    需要在不同类型的数据之间建立联系，并识别哪一类数据决定“订阅户总数”（即绝对数 = 总数 × 比例）。